Taha Bekar

Robotics Software Engineer — Perception — SLAM — Autonomous Systems
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Education

Rensselaer Polytechnic Institute Troy, NY
M.S. Electrical, Computer, and Systems Engineering
2024 – May 2026
Focus: Autonomy, Robotics, and Control

Middle East Technical University Ankara, Turkey
B.S. Electrical and Electronics Engineering
2018 – 2024
GPA: 3.56 / 4.00

Research Experience

Deformable Object Manipulation Research – RPI 2025 – Present
Research Assistant / Master's Project

- Developed a Python-based, ROS2-compatible deformable-object simulation framework in NVIDIA Isaac Sim using Newton Style3D for real-time cloth manipulation with anisotropic material, contact, friction, damping, and bending parameters.
- Built an obstacle-aware cloth manipulation pipeline in Python with kinematic corner pinning, simulated cloth state publishing, ROS2 command interfaces, obstacle-distance computation, and closed-loop follower-corner control.
- Integrated a Control Barrier Function safety controller in Python for cloth-obstacle avoidance in contact-rich manipulation scenarios.
- Developed a real-time perception and state estimation pipeline in Python for planar deformable objects using RGB-D camera data, ArUco marker tracking, and Kalman filtering in ROS for real-to-sim calibration.
- Developing multi-robot manipulation for deformable objects in NVIDIA Isaac Sim, with coordinated motion and internal tension control, and using NVIDIA Isaac Lab to train Reinforcement Learning controllers.

Work Experience

Kuartis Ankara, Turkey
Computer Vision / Machine Learning Engineer Feb 2024 – Jun 2024

- Developed and benchmarked real-time, production-style C++ implementations of LiDAR point cloud registration algorithms, including ICP, NDT, GICP, and FAST-GICP, for SLAM and mapping systems.
- Implemented real-time LiDAR–IMU sensor fusion pipelines in C++ by combining LiDAR odometry with IMU odometry and evaluating tightly coupled and loosely coupled fusion strategies for improved localization robustness.
- Evaluated C++ implementations of SLAM frameworks including LOAM, LIO-SAM, and FAST-LIO for localization accuracy, runtime performance, and computational efficiency.
- Improved LIO-SAM localization accuracy by 6% by integrating GICP-based scan matching in C++ into the GTSAM factor-graph optimization framework.

GT-ARC Gemeinnützige GmbH, DAI Labor, TU Berlin Berlin, Germany
Software Engineer Intern Jul 2023 – Sep 2023

- Developed a production-oriented C++ LiDAR-based free parking space detection algorithm for autonomous vehicles using ROS, Autoware, and HD Maps.
- Implemented geometric intersection analysis in C++ between detected vehicles and mapped parking regions, achieving 86% free space detection accuracy.

Analog Devices Istanbul, Turkey
Software Engineer Intern Jul 2022 – Aug 2022

- Developed production-level C firmware for a PWM controller on the MAX32666 microcontroller using hardware timer peripherals; the implementation was incorporated into the official microcontroller firmware.
- Built a GUI interface for validating PWM signals and implemented automated unit tests using Google Test for the production firmware.

Projects

Autonomous UGV Mapping System

- Converted a 2D LiDAR into a 3D scanning system via servo rotation to enable volumetric mapping.
- Implemented OctoMap mapping using ICP (Iterative Closest Point) registration and AMCL (Adaptive Monte Carlo Localization) algorithms in ROS. Achieved accuracy levels of 95% and 75% for mapping and localization, respectively.

Lane Detection System

- Implemented classical vision-based lane detection pipelines in C++ and MATLAB using Canny edge detection and the Hough transform.

Technical Skills

Programming: Python, C++17, C, MATLAB
Robotics: ROS, ROS2, NVIDIA Isaac Sim, NVIDIA Isaac Lab, PhysX, Newton, Gazebo, 3D Kinematics, SLAM, Reinforcement Learning
Perception: RGB-D Perception, LiDAR Point Cloud Processing, Sensor Fusion, ROS Bags, TF, OptiTrack, Computer Vision
Libraries: PCL, OpenCV, PyTorch, NumPy, rclcpp, tf2, Eigen, GTSAM, Google Test
Tools: Linux, Git, Docker